

The enhancement of Passenger Ground Transfer Management by the BPMN modeling and Real-Time Flight Data

Giorgos Mpardis
Department of Digital Systems
University of the Peloponnese
Sparta, Greece
0009-0003-4209-317X

Dionisis Margaritis
Department of Digital Systems
University of the Peloponnese
Sparta, Greece
0000-0002-7487-374X

Konstantinos I. Roumeliotis
Department of Informatics and
Telecommunications
University of the Peloponnese
Tripoli, Greece
0000-0002-8098-1616

Costas Vassilakis
Department of Informatics and
Telecommunications
University of the Peloponnese
Tripoli, Greece
0000-0001-9940-1821

Dimitris Spiliotopoulos
Department of Management Science
and Technology
University of the Peloponnese
Tripoli, Greece
0000-0003-3646-1362

Abstract—Passenger ground transfer involves managing the movement of fleets of vehicles to shuttle passengers between airports and hotels. A persistent challenge and source of operational inefficiency in this environment is deadheading - driving an empty vehicle to pick up passengers. In this paper, we describe a Decision Support System (DSS) to optimally assign transfer routes. The DSS is modeled using BPMN 2.0 and uses real-time flight data, regulatory constraints, and geo-specific information to minimize the number of deadheading kilometers driven. The decision engine provides optimized driver assignment suggestions (ordered by the best option), which the dispatchers can then accept or reject. We validated our system with a real-world dataset of 987,342 transfers collected from a commercial service operating 734 transfer companies in Greece, Cyprus, and Belgium, where 3,700 vehicles and 5,400 drivers were registered. We found that the number of deadheading kilometers per transfer was reduced by 23.8%, the transfer cost decreased by 23.8%, driver utilization increased by 7.2 percentage points, and customer waiting times for transfers in the case of a flight delay decreased by 45.2%. The use of the DSS also reduced the dispatcher workload (on average by an estimated 5.1 hours per day) by offering automated suggestions.

Keywords—Business Process Modeling, BPMN, Decision Support System, Route Optimization, Passenger Ground Transfer, Deadheading Minimization, Real-Time Flight Data, Key Performance Indicators

I. INTRODUCTION

Passenger ground transfer companies transport millions of passengers per year between airports and their destinations. These companies use vehicles that are allocated to drivers by dispatchers in the company offices. The allocation of vehicles to drivers directly affects operational cost, the number of transfers completed within each driver's shift, driver productivity, and customer satisfaction [1], [2], [3].

The number of vehicles that have to make 'empty' transits (i.e., without passengers or cargo) to reach a pickup location is a major issue, since such transits lead to fuel consumption,

cause vehicle wear, and furthermore make no contribution to drivers' completed transfer count. These empty vehicle transits are referred to as 'deadheading' [4], [5], [6].

Despite the scale and economic importance of ground transfer operations, the deadheading problem has received limited attention in the business process management and optimization literature. Some researchers in the domain of vehicle routing problems have tackled issues intersecting with the deadheading problem, such as the pickup and delivery problem [7], [8], [9] or the scheduling of airport shuttle transfers [10], [11], [12].

However, the approaches in the existing literature adopt a fragmented approach, examining partial issues in isolation, and do not place decisions in a formally modeled, holistic business process. An important omission relates to the fact that real-time data streams, such as real-time flight information, are either not used at all in the assignment procedure or are underutilized, despite flight delays being a recurring cause of disruptions to airport ground transfers [13], [14], [15].

Although researchers have recognized the benefit of BPMN-based process modeling for logistics [16], [17], [18], and a Key Performance Indicator (KPI)-based process optimization [19], [20], [21], none of the existing approaches combines formal process modeling, multi-parameter optimization, and a real-time data integration for airport ground transfers.

In this paper, we present a DSS for the optimization of ground transfer route assignment. The proposed DSS is formally modeled using BPMN (Business Process Model and Notation), and takes into consideration real-time flight data, the regulatory requirement of 30-minute inter-transfer period, and geographical data for minimizing deadheading and improving the quality of service. The DSS includes an optimization tool that formulates ranked proposals for drivers, and these proposals can be accepted or modified by the dispatcher. We evaluate the efficacy of the proposed approach

on a real-world dataset sourced from a commercial platform that handles 987,342 transfers in three countries. The evaluation involves six KPIs (Average Deadheading Distance, Total Deadheading Distance, Cost per Transfer, Driver Utilization Rate, Customer Wait Time, and Dispatcher Assignment Time) and the results have asserted that the proposed approach achieves considerable improvements for each of the KPIs.

The rest of the paper is organized as follows: Section II overviews related work, whereas in Section III the system architecture and the BPMN model are introduced. Section IV presents the evaluation procedure and the results, and Section V provides a discussion. Finally, Section VI concludes the paper and outlines future work.

II. RELATED WORK

A. BPMN in Transportation and Logistics

BPMN has been employed to model and analyze transportation and logistics processes. The work in [22] modeled the production logistics system of the automotive industry using BPMN 2.0. This work asserts that complex, multi-actor business processes can be modeled modularly with BPMN. The work in [23] proposed BPMN-based models for the container transportation process, complemented by a method that maps BPMN models to the ACTIF reference architecture (Aide à la Conception de Systèmes de Transports Interopérables en France — the French national framework architecture for interoperable intelligent transport systems) for intermodal transport. The study in [16] used the MIPI (Model-based and Integrated Process Improvement) methodology combined with BPMN for process analysis, evaluated activities using value-added and cause-and-effect analysis, identified delays and problems, and proposed a redesign aimed at reducing non-value-added activities and implementing parallel processes to improve efficiency. The work in [24] used BPMN 2.0 to map and analyze logistics processes in SMEs, providing insights into functionality, monitoring, improvement, automation, and support for analysts, IT professionals, and business process managers.

The above works demonstrate that BPMN is a prominent modeling language for the transportation process. However, none of these works deals with the ground transfer process, nor with the integration of external real-time information into the modeled process.

B. Vehicle Routing and Airport Transfer Optimization

The Vehicle Routing Problem (VRP) and its variants provide the algorithmic foundation for transport optimization.

The work in [4] proposed the VRP with Pickup and Delivery and Time Windows, which defines the scope of problems when vehicles are assigned tasks of transporting people from their origins to their destinations on or before the assigned time. The study in [7] provided a survey of heuristic and metaheuristic methods applied to the Pickup and Delivery Problem, such as genetic algorithms, simulated annealing, and variable neighborhood search. The work in [25] presented the FSPD algorithm for the VRP with Simultaneous Pickup and Delivery (VRPSPD) and the VRP with Mixed Pickup and Delivery (VRPMPD) — variants in which each customer requires both a pickup and a delivery, or either one, respectively — extending the FILO (Fast Iterated Local Search Localized Optimization) framework. This approach achieved competitiveness with modern models, efficient

solution of large-scale instances, linear scaling of computation time, and reliable management of multiple pickup and drop-off points.

Regarding the airport field, the work in [26] presented a VRP model for airport customer pick-up and delivery service with consideration of carbon emission costs. The study in [10] elaborated on the optimization of airport shuttle routes at the Dallas Fort Worth International Airport using a Static Dial VRP, which attained a 9% decrease in fuel costs.

The above-mentioned studies dealt with important aspects of vehicle routing optimization in airport environments, yet in an isolated fashion where each problem is tackled individually, without considering the broader business process within which routing decisions are made, and do not consider real-time flight data.

C. Decision Support Systems and Real-Time Dispatch

The work in [13] developed a taxonomy of dynamic and stochastic VRPs and argued that real-time integration of information makes routing more effective, although it increases computational complexity. The study in [27] proposed a digital twin-based approach for scheduling specialized airport vehicles, adapting schedules to changing conditions, through real-time monitoring.

Flight delays affect a substantial proportion of commercial flights, directly impacting passenger arrival times and, consequently, optimal ground transfer scheduling. Despite this fact, the use of real-time flight status data in ground transfer route optimization has not been addressed in the literature.

D. KPI-Based Business Process Optimization

The study in [19] proposed an extension of the BPMN meta-model, incorporating KPI definitions at multiple process levels for automated performance evaluation. The work in [28] combined process mining with preconfigured KPI targets and redesign patterns to drive process improvement. The work in [29] provided a comprehensive treatment of the BPM lifecycle, covering process discovery, analysis, redesign, and monitoring through performance indicators.

These frameworks provide the methodological basis for KPI-driven process improvement, but have not been applied to ground transfer operations, where domain-specific KPIs, such as deadheading distance and flight-delay-adjusted wait times, are central.

III. SYSTEM ARCHITECTURE AND BPMN MODEL

A. System Overview

This work introduces a commercial transfer management system for businesses operating in the domain of airport ground transfer. The solution targets transfer companies that operate a vehicle fleet and conduct airport to hotel transfer business, which is a point-to-point transport service. The system consists of five components as follows:

1. The Booking engine, which processes and manages the transfer bookings;
2. The Dispatch system, which facilitates the task of the dispatcher to assign each transfer to a specific driver;
3. The Driver mobile application, which provides the driver with notifications regarding their assignments and guides the driver by providing navigation;

4. The Flight tracker system, which monitors the live status information from the airline's flight status feeds; and
5. The Optimization engine, which computes and ranks the possible assignments between a driver and a transfer.

Fig. 1 illustrates the system architecture. These transfer bookings are received through multiple channels and are collected in a central database; the optimization engine considers the pending transfers and the available drivers under the various geo-temporal and legal constraints. Recommendations are given to the dispatcher who can either approve or modify them.

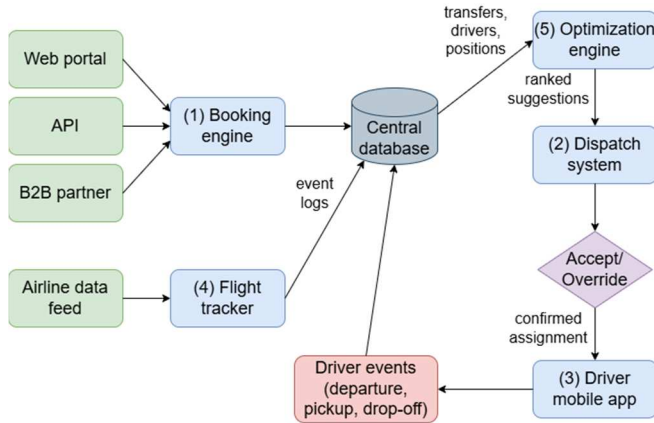


Fig. 1. Architecture of the proposed transfer management and optimization platform.

B. BPMN Process Model

The transfer assignment process was modeled using BPMN 2.0. Fig. 2 presents the proposed process model, which incorporates the optimization engine into the assignment workflow.

The process starts when a new transfer booking enters the system. The first activity retrieves the real-time flight status from the airline data feed. A decision gateway evaluates the flight status: if the flight is delayed, the system adjusts the estimated pickup time; if the flight is cancelled, the transfer

is flagged and the process terminates with a customer notification.

For confirmed arrivals, the optimization engine is triggered. It retrieves available drivers and their current positions (which are stored in the company's database, and are continuously updated upon reception of driver events), computes a multi-parameter score for each candidate driver, and ranks the results. The top-ranked suggestions are presented to the dispatcher as an ordered list. The dispatcher reviews the suggestions and either accepts or modifies the assignment. Upon confirmation, the assigned driver receives a notification through the mobile application with the pickup location, estimated pickup time, and any flight delay information. Throughout the execution phase, driver events (departure, arrival at pickup, customer pickup, and drop-off) are recorded, enabling the post-hoc KPI computation.

C. Optimization Parameters

Five parameters that are input to the optimization engine are:

1. Company Origin: Each transfer company has a designated company origin location where deadheading is calculated from when there is no preceding transfer for the assigned driver.
2. Pickup Locations: The coordinates of the pickup locations (i.e., terminal or hotel address) of the outstanding transfers. This is used to calculate the traveling distance from all candidate drivers' locations.
3. Driver Position: The last reported location of each candidate driver (from their most recent completed transfer) or the company origin when no preceding transfer exists. The optimization engine uses driver positions together with the pickup location to compute the deadheading distance — the empty-vehicle travel distance that the system seeks to minimize.
4. Legal Constraint (30-Minute Rule): A 30-minute window between consecutive transfer assignments must be honored according to regulations; this is a hard constraint used for identifying candidate drivers.
5. Real-time flight status: It is possible to obtain information on each flight's scheduled time, scheduled

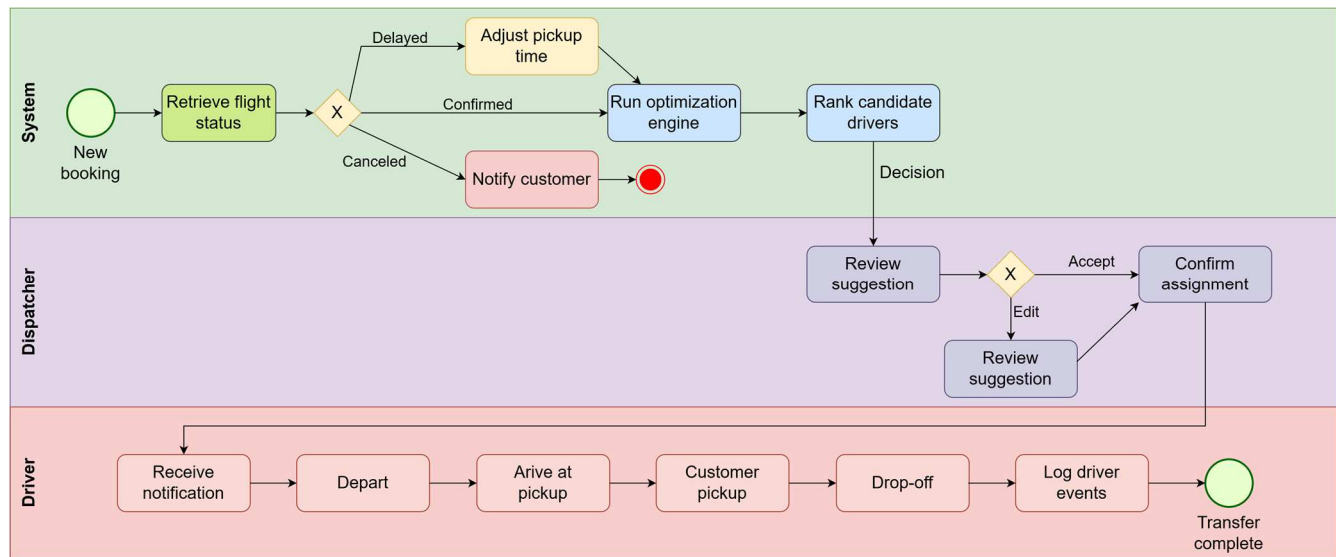


Fig. 2. BPMN 2.0 model of the transfer assignment process with optimization engine integration.

arrival time, and whether it is delayed by the airlines. Real-time flight status is utilized in two ways: (a) to adjust the estimated pickup time, in order to minimize drivers' waiting time on delayed flights, and (b) to trigger a re-assignment process when a change in the flight schedule (e.g., a delay or cancellation) makes the current driver-transfer assignment infeasible or sub-optimal — for instance, when the revised pickup time would breach the 30-minute rule for the already-assigned driver, or would make a closer driver available.

Other regulatory constraints — driving-time and rest limits, vehicle-category and passenger-capacity rules, and driver licensing — are enforced upstream at shift planning, so every candidate driver is ensured to be compliant. Only the 30-minute rule is modelled explicitly, since it depends only on the driver's specific transfer sequence and must be checked dynamically at assignment time.

D. Optimization Algorithm

The route assignment task can be stated as an optimization problem. Let $T = \{t_1, t_2, \dots, t_n\}$ be the set of pending transfers and $D = \{d_1, d_2, \dots, d_m\}$ be the set of available drivers. Each driver d_j has a known position $pos(d_j)$ and the last timestamp $lstamp(d_j)$ of their most recent completed transfer. Each transfer t_i has a pickup location $loc(t_i)$ and a scheduled pickup time $tpickup(t_i)$. The objective is to find an assignment $\sigma: T \rightarrow D$ that minimizes the total deadheading distance, i.e., the quantity:

$$\min \sum_{t_i \in T} distance(pos(\sigma(t_i)), loc(t_i)) \quad (1)$$

subject to the following constraints:

1. $tpickup(t_i) - lstamp(\sigma(t_i)) \geq 30 \text{ min}, \forall t_i: \sigma(t_i) \neq \text{NULL}$
2. Each driver is assigned at most one pending transfer.

This is an instance of the minimum-weight bipartite matching problem with side constraints. Given the real-time nature of operations and moderate instance sizes, we adopt a greedy heuristic approach to compute the solution. In more detail, the optimization procedure applies a greedy heuristic whenever a new transfer has to be assigned or when a reassignment is needed following a change in flight status, according to the following algorithm:

Input: Pending transfers $T = \{t_1, t_2, \dots, t_n\}$, available drivers $D = \{d_1, d_2, \dots, d_m\}$, driver positions P , and flight status data F .

- **Step 1 — Flight Status Update:** For every pending transfer t of an arriving flight, get the status of the flight. If it is delayed, then update the pickup time.
- **Step 2 — Constraint Filtering:** Exclude drivers violating the 30-minute rule. Driver d_j is eligible for transfer t_i only if $tpickup(t_i) - lstamp(d_j) \geq 30 \text{ min}$.
- **Step 3 — Deadheading Distance Computation:** For each eligible driver d_j , compute $DH(d_j, t_i) = dist(p_j, loc_i)$.
- **Step 4 — Ranking:** Sort the qualifying drivers from smallest DH to largest DH. The driver with the

shortest distance traveled with an empty vehicle is suggested first.

- **Step 5 — Presentation:** Present the top-ranked driver suggestions to the dispatcher. The dispatcher may accept the highest-ranked option, select an alternative from the list, or manually assign a different driver.

The algorithm is event-driven, reacting to the arrival of new booking requests, updates on flight information, and the status of available drivers. The time complexity of the algorithm is $O(n*m)$ with n pending transfers and m drivers available, which is appropriate for real-time usage given the usual scale of operations.

The model assumes that (a) each driver's position is known (the drop-off of their last transfer, or the company origin), (b) deadheading is approximated by geographic distance, (c) the 30-minute rule is a hard constraint whereas other regulations are satisfied upstream, and (d) each run assigns at most one pending transfer per driver, optimized greedily rather than globally. Transfer times are treated as deterministic — derived from these distances and the delay-adjusted pickup times — with stochastic variability (traffic, passenger processing) left for future work (Section VI).

IV. EVALUATION

A. Dataset Description

The system was tested on actual data from an existing commercial transfer management platform. The dataset consists of 987,342 transfer records that were collected over 36 months, from January 2023 to December 2025. The platform is active in 3 countries, Greece, Cyprus, and Belgium, used by 734 transfer companies, 3700 connected vehicles, and 5400 registered drivers. The characteristics of the dataset are summarized in Table I.

A single transfer record comprises: (a) the booking timestamp, (b) the coordinates of the pickup and drop-off locations, (c) the planned and actual pickup times, (d) the booked driver, (e) the flight number (only for airport pickups), (f) the actual arrival time of flight, and (g) the information about events that occurred during the specific transfer, with each element consisting of a timestamp and a description. Field (f) indicates the time instant the aircraft lands (from the flight feed), whereas field (c) corresponds to the instant when the driver collects the passenger; the two differ by disembarkation, immigration, and baggage-claim time (typically 20–45 min) — the window used to adjust pickups to delayed arrivals.

TABLE I. DATASET SUMMARY

Metric	Value
Total transfers	987,342
Date range	January 2023 – December 2025
Transfer companies	734
Connected vehicles	3,700
Registered drivers	5,400
Active countries	3 (Greece, Cyprus, Belgium)
Avg. transfers per day	912

B. KPI Definitions

Six KPIs were identified and used to assess the optimization engine:

1. **Average Deadheading Distance (ADD):** This is defined as the average empty travel distance per transfer.
2. **Total Deadheading Distance (TDD):** This is calculated as the total sum of empty travel distance across all transfers.
3. **Cost per Transfer (CPT):** This is estimated as the proportion of driver operational cost due to deadheading, dependent on travel distance and time.
4. **Driver Utilization Rate (DUR):** This is defined as the proportion of active travel time to total working time, measured in percentage.
5. **Customer Wait Time (CWT):** This is computed as the difference between actual driver arrival and scheduled pickup time, measured in minutes.
6. **Dispatcher Assignment Time (DAT):** This is calculated as the time between a transfer entering the assignment queue to it being assigned to a driver, measured in minutes.

C. Experimental Setup

In our experimental setup, two groups of transfers were compared:

- **Optimized group:** Transfers where the dispatcher accepted the system's suggestion (775,849 transfers, 78.6% of total cases).
- **Baseline group:** Transfers where the dispatcher overrode the suggestion and manually assigned a different driver (211,493 transfers, 21.4% of total cases).

Group membership is read directly from the operational logs: the platform stores both the engine's ranked suggestions and the driver finally assigned by the dispatcher, so a transfer is considered "Optimized" when these match and "Baseline" when the dispatcher overrode it. The override is the only difference between the groups, and is the factor that introduces the extra deadheading; the reasons for overriding are not recorded, a selection-bias caveat that we revisit in Subsection V.C.

This experimental design uses the dispatcher's decision as a quasi-experimental condition. For transfers involving arriving flights, we separately evaluated the effect of real-time flight data integration by comparing customer wait times for transfers where flight delays were detected and pickup times were automatically adjusted versus transfers without adjustment.

D. Results

This subsection presents the results of the evaluation in relation to the six KPIs listed in Subsection IV.A, confidence intervals (CI) and standard deviations (SD) are listed where appropriate.

1. **Average Deadheading Distance:** Optimized transfers had a mean deadheading distance of 8.43 km (95% CI: 8.40–8.46, SD = 4.7), compared to 11.2 km (95% CI: 11.13–11.27, SD = 6.3) for manually assigned transfers — a 23.8% reduction ($p < 0.001$). Fig. 3 illustrates the distribution of deadheading distances (km) for optimized vs. baseline transfers.
2. **Total Deadheading Distance:** Regarding the Total Deadheading Distance KPI, extrapolating each group

(optimized, baseline) to the total number of transfers, we estimate that the total distance for the optimized group would be 8.42 million km, while the total distance for the baseline group would be 11.06 million km, i.e., the optimized group exhibits savings of 23.84%.

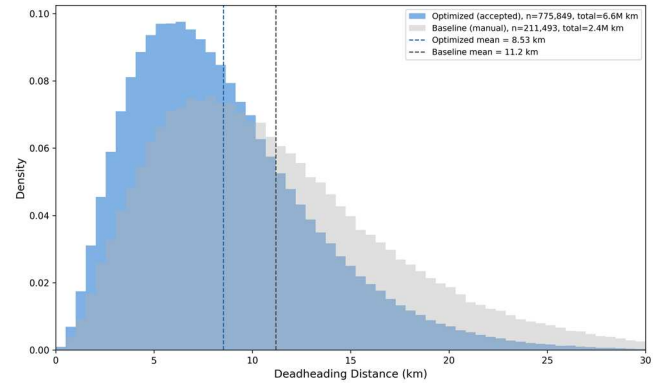


Fig. 3. Distribution of deadheading distances (km) for optimized vs. baseline transfers.

3. **Cost per Transfer:** The estimated deadheading cost was 3.84 EUR (95% CI: 3.82–3.86) per transfer in the optimized group, versus 5.04 EUR (95% CI: 4.98–5.10) in the baseline group, accounting for a reduction equal to 23.81%. Over the 36-month period, this amounts to approximately 935,700 EUR in savings across the platform.
4. **Driver Utilization:** Driver utilization was 74.3% (95% CI: 74.2–74.4) under optimized assignments versus 67.1% (95% CI: 66.9–67.3) for manual assignments, corresponding to an improvement of 7.2 percentage points (+7.2 pp). Fig. 4 depicts the performance improvement achieved by the proposed system regarding the driver utilization metric.

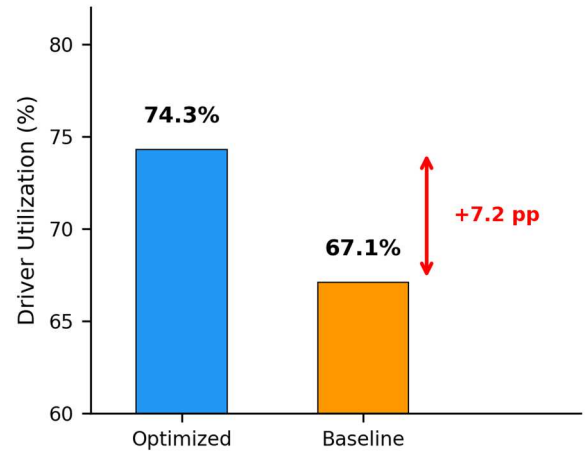


Fig. 4. Driver utilization rate comparison between optimized and baseline assignments.

5. **Customer Wait Time:** For airport pickup trips where a flight delay was detected (31.4% of all airport transfers), the pickup time was automatically adjusted. The average wait time for customers was 6.8 minutes (95% CI: 6.7–6.9) (with adjustment) vs. 12.4 minutes (95% CI: 12.2–12.6) (without adjustment), accounting for an improvement of 45.2%.

6. *Dispatcher Assignment Time*: Dispatchers had a decrease in time spent per transfer from 3.2 minutes (review manually) to 0.7 minutes (review & confirm). Since there are 912 trips per day, this represented a savings of 5.1 cumulative dispatcher hours per day. This metric was derived from aggregated operational logs rather than per-transfer timing measurements; therefore, a statistical significance test was not applicable.

Table II summarizes the KPIs evaluated in this study.

V. DISCUSSION

A. Findings

The main strengths of the approach are that it embeds the routing decision in a formally modeled, end-to-end BPMN process, integrates real-time flight data, and keeps the dispatcher in control through ranked, overridable suggestions (78.6% acceptance). On the other hand, its main weaknesses can be traced to the use of a single-platform dataset, the quasi-experimental design, the greedy per-transfer heuristic, and geographic rather than traffic-aware distances. Weaknesses are discussed in more detail in Subsection V.C.

The reduction of 23.8% observed in the distance drivers spend deadheading affects the cost per transfer and the driver utilization, which are included in the KPIs, but also leads to decreased fuel expenditures and diminished wear on drivers' vehicles; a cost savings estimated to be about 311,000 EUR per year. The positive impact on environmental pollution reduction is also an important side effect of this optimization process. The dispatcher acceptance rate of 78.6% is crucial as it suggests that the operating employees find the generated suggestions useful and accept their usage, which would be a prerequisite for the implementation and adoption of any DSS. By providing ranked driver suggestions that dispatchers could accept with a single action, the system reduced dispatcher assignment effort by an estimated 5.1 hours per day, freeing time for exception handling, customer communication, and quality assurance.

The integration of real-time flight data reduced customer wait times by 45.2% for delayed flights. By automatically adjusting pickup times to match actual arrival times, the system prevents premature driver dispatch — thereby eliminating idle waiting at the airport and freeing up drivers for revenue-generating assignments.

B. Comparison with Related Work

The proposed system integrates the optimization problem within the process modeled with a formal language, thus adopting a holistic view of the business operations that integrates all involved processes and exploits the available data sources to achieve an optimized result. This distinguishes the proposed approach from previous works, which model airport vehicle routing as an algorithmic problem [10], [26], thereby optimizing just a fragment of the process.

The assignment business process modeled by BPMN not only facilitates communication between stakeholders but also enables systematic identification of bottlenecks that may occur within the process. Thus, we extend the KPI-based process improvement approach presented in [19], [28] to domain-specific optimization for ground transfers.

The size of the test case (987,342 transfers across three countries) is considerably larger than most of those presented in the literature on the same topic [10], [26], [27]. This provides increased warranties concerning the generalizability of the findings.

C. Limitations

While the proposed research overcomes limitations of previous research, a number of limitations still persists. First, the evaluation uses data from a single platform, which may limit generalizability. Second, the quasi-experimental design —comparing dispatcher-accepted versus dispatcher-overridden assignments— may introduce selection bias, since dispatchers may override suggestions for reasons not captured in the data (e.g., last-minute availability changes, customer preferences). Third, the greedy heuristic optimizes each transfer independently rather than performing global optimization across pending transfers, leaving room for more sophisticated algorithmic approaches. Fourth, deadheading distance computation uses geographic distance rather than real-time traffic-aware routing, which may underestimate actual travel times in congested areas. These limitations will form the primary basis for our future work.

D. Practical Implications

The decrease in mileage of return trips helps save fuel, thus reducing the CO₂ emissions and meeting economic and environmental objectives. The DSS design (suggestion system with dispatcher decision), which balances optimality

TABLE II. KPI COMPARISON — OPTIMIZED VS. BASELINE

<i>KPI</i>	<i>Baseline</i>	<i>Optimized</i>	<i>Improvement</i>	<i>p-value</i>
Avg. Deadheading (km)	11.2	8.53	23.8%	< 0.001
Total Deadheading (km, extrapolated)	11.05 M	8.42 M	23.84%	-
Cost/Transfer (EUR)	5.04	3.84	23.81%	< 0.001
Driver Utilization (%)	67.1	74.3	+7.2 pp	< 0.001
Customer Wait (min)	12.4	6.8	45.2%	< 0.001
Dispatcher Assignment Time (min, estimated)	3.2	0.7	78.1%	—
Dispatcher Acceptance	—	78.6%	—	—

and practical applicability can be integrated into existing systems to enhance their usage and effectiveness. The BPMN model can serve as an anchor to guide the auditing process, training of employees, as well as documentation for the system's regulatory and legal requirements.

VI. CONCLUSION AND FUTURE WORK

In this paper, we presented a DSS for assigning the best route for ground transfer, which combines real-time flight information, regulatory constraints, and geo-information to reduce deadheading and increase service quality. The proposed DSS has been formally modeled using BPMN 2.0, and its implementation encompasses a greedy optimization algorithm, which determines the assignments of drivers to ground transfers, catering for deadheading minimization and ensuring compliance with the regulatory requirement of a 30-minute inter-transfer constraint.

The system was evaluated on 987,342 transfers from a real commercial platform covering 3 countries (734 companies, 3,700 vehicles and 5,400 drivers). The evaluation results showed significant improvements in multiple metrics, including a 23.8% decrease in average deadheading distance, a 23.8% decrease in cost per transfer, a 7.2 percentage point increase in driver utilization and a 45.2% decrease in customer waiting time in the case of delays. The workload for dispatchers is estimated to be reduced by 5.1 hours per day.

In our future work, we will investigate machine learning methods for predicting demand in order to optimize the waiting locations of vehicles and drivers, incorporate real-time traffic data in order to more accurately estimate deadheading distances, and develop new dynamic re-assignment policies.

REFERENCES

- [1] O. J. Ibarra-Rojas, F. Delgado, R. Giesen, and J. C. Muñoz, "Planning, operation, and control of bus transport systems: A literature review," *Transportation Research Part B: Methodological*, vol. 77, pp. 38–75, Jul. 2015, doi: 10.1016/j.trb.2015.03.002.
- [2] P. Ammann, R. Kolisch, and M. Schiffer, "Driver routing and scheduling with synchronization constraints," *Transportation Research Part B: Methodological*, vol. 174, p. 102772, Aug. 2023, doi: 10.1016/j.trb.2023.05.009.
- [3] D. Hörcher and A. Tirachini, "A review of public transport economics," *Economics of Transportation*, vol. 25, p. 100196, Mar. 2021, doi: 10.1016/j.ecotra.2021.100196.
- [4] P. Toth and D. Vigo, Eds., *The Vehicle Routing Problem*. Society for Industrial and Applied Mathematics, 2002. doi: 10.1137/1.9780898718515.
- [5] O. Tengilimoglu, O. Carsten, and Z. Wadud, "Implications of automated vehicles for physical road environment: A comprehensive review," *Transportation Research Part E: Logistics and Transportation Review*, vol. 169, p. 102989, Jan. 2023, doi: 10.1016/j.tre.2022.102989.
- [6] S. Islam, Y. Shi, J. U. Ahmed, and M. J. Uddin, "Minimization of empty container truck trips: insights into truck-sharing constraints," *IJLM*, vol. 30, no. 2, pp. 641–662, May 2019, doi: 10.1108/IJLM-08-2018-0191.
- [7] S. N. Parragh, K. F. Doerner, and R. F. Hartl, "A survey on pickup and delivery problems: Part II: Transportation between pickup and delivery locations," *Journal für Betriebswirtschaft*, vol. 58, no. 2, pp. 81–117, Jun. 2008, doi: 10.1007/s11301-008-0036-4.
- [8] D. Dumez, F. Lehuédé, and O. Péton, "A large neighborhood search approach to the vehicle routing problem with delivery options," *Transportation Research Part B: Methodological*, vol. 144, pp. 103–132, Feb. 2021, doi: 10.1016/j.trb.2020.11.012.
- [9] V. F. Yu, H. Susanto, Y.-H. Yeh, S.-W. Lin, and Y.-T. Huang, "The Vehicle Routing Problem with Simultaneous Pickup and Delivery and Parcel Lockers," *Mathematics*, vol. 10, no. 6, p. 920, Mar. 2022, doi: 10.3390/math10060920.
- [10] D. Sigler *et al.*, "Route optimization for energy efficient airport shuttle operations – A case study from Dallas Fort worth International Airport," *Journal of Air Transport Management*, vol. 94, p. 102077, Jul. 2021, doi: 10.1016/j.jairtraman.2021.102077.
- [11] Z. Cao and A. (Avi) Ceder, "Autonomous shuttle bus service timetabling and vehicle scheduling using skip-stop tactic," *Transportation Research Part C: Emerging Technologies*, vol. 102, pp. 370–395, May 2019, doi: 10.1016/j.trc.2019.03.018.
- [12] K. Alamatsaz, F. Quesnel, and U. Eicker, "Enhancing Electric Shuttle Bus Efficiency: A Case Study on Timetabling and Scheduling Optimization," *Energies*, vol. 17, no. 13, p. 3149, Jun. 2024, doi: 10.3390/en17133149.
- [13] V. Pillac, M. Gendreau, C. Guéret, and A. L. Medaglia, "A review of dynamic vehicle routing problems," *European Journal of Operational Research*, vol. 225, no. 1, pp. 1–11, Feb. 2013, doi: 10.1016/j.ejor.2012.08.015.
- [14] E. Boumpa *et al.*, "A Review of the Vehicle Routing Problem and the Current Routing Services in Smart Cities," *Analytics*, vol. 2, no. 1, pp. 1–16, Dec. 2022, doi: 10.3390/analytics2010001.
- [15] L. Abbatecola, M. P. Fanti, and W. Ukovich, "A review of new approaches for Dynamic Vehicle Routing Problem," in *2016 IEEE International Conference on Automation Science and Engineering (CASE)*, Fort Worth, TX, USA: IEEE, Aug. 2016, pp. 361–366. doi: 10.1109/COASE.2016.7743429.
- [16] M. Wuennenberg, B. Wegerich, and J. Fottner, "Optimization of Internal Logistics using a Combined BPMN and Simulation Approach," in *ECMS 2022 Proceedings edited by Ibrahim A. Hameed, Agus Hasan, Saleh Abdel-Afou Alaliyat*, ECMS, Jun. 2022, pp. 13–19. doi: 10.7148/2022-0013.
- [17] M. Stajniak and B. Guszczak, "Analysis of Logistics Processes According to BPMN Methodology," in *Information Technologies in Environmental Engineering*, vol. 3, P. Golinska, M. Fertsch, and J. Marx-Gómez, Eds., in Environmental Science and Engineering, vol. 3, Berlin, Heidelberg: Springer Berlin Heidelberg, 2011, pp. 537–549. doi: 10.1007/978-3-642-19536-5_42.
- [18] N. Bandaranayake, S. Kiridena, and A. K. Kulatunga, "Capturing cross-border logistics for analysis and improvement," *JGOSS*, vol. 17, no. 2, pp. 186–208, Apr. 2024, doi: 10.1108/JGOSS-05-2022-0037.
- [19] A. Wannes and S. A. Ghannouchi, "KPI-Based Approach for Business Process Improvement," *Procedia Computer Science*, vol. 164, pp. 265–270, 2019, doi: 10.1016/j.procs.2019.12.182.
- [20] M. Piechowski and A. Meller, "Use of KPI Indicators to Optimize Production Process Performance – Case Study," in *Distributed Computing and Artificial Intelligence, Special Sessions I, 20th International Conference*, vol. 741, R. Mehmood, V. Alves, I. Praça, J. Wikarek, J. Parra-Domínguez, R. Loukanova, I. De Miguel, T. Pinto, R. Nunes, and M. Ricca, Eds., in Lecture Notes in Networks and Systems, vol. 741, Cham: Springer Nature Switzerland, 2023, pp. 3–12. doi: 10.1007/978-3-031-38318-2_1.
- [21] T. B. Sohrab, "Predictive Analytics and KPI Dashboards for Enterprise Workflow Optimization," *asrc*, vol. 01, no. 01, pp. 1848–1886, Jan. 2025, doi: 10.63125/5x0d1702.
- [22] Khabbazi, M. R., Hasan, M. K., Sulaiman, R., and Shapi'i, A., "Business Process Modelling in Production Logistics: Complementary Use of BPMN and UML," *Middle East Journal of Scientific Research*, vol. 15, no. 4, pp. 516–529, 2013.
- [23] Hadj-Khalifa, H. and Kamel, M. A., "Transportation process of containers BPMN-modeling and transformation into ACTIF model," *Romanian Journal of Information Science and Technology*, vol. 18, no. 2, pp. 77–93, 2015.
- [24] R. Sulaiman, M. N. Ahmad, M. Khabbazi, and M. A. Riza, "Outbound logistics business process modeling: Analytic perspective with BPMN 2.0," in *Uncertainty in Computational Intelligence-Based Decision Making*, Elsevier, 2025, pp. 23–54. doi: 10.1016/B978-0-443-21475-2.00013-8.
- [25] F. Cavaliere, L. Accorsi, D. Laganà, R. Musmanno, and D. Vigo, "An efficient heuristic for very large-scale vehicle routing problems with simultaneous pickup and delivery," *Transportation Research Part E: Logistics and Transportation Review*, vol. 186, p. 103550, Jun. 2024, doi: 10.1016/j.tre.2024.103550.
- [26] Y. Peiyang, T. Jiafu, and Y. Yang, "Based on low carbon emissions cost model and algorithm for vehicle routing and scheduling in picking up and delivering customers to airport service," in *2013 25th Chinese Control and Decision Conference (CCDC)*, Guiyang, China: IEEE, May 2013, pp. 1693–1697. doi: 10.1109/CCDC.2013.6561203.

- [27] Q. Luo, H. Liu, C. Liu, and Q. Deng, "Multi-strategy cooperative scheduling for airport specialized vehicles based on digital twins," *Sci Rep*, vol. 14, no. 1, p. 15533, Jul. 2024, doi: 10.1038/s41598-024-66350-0.
- [28] J. Cherni, R. Martinho, and S. A. Ghannouchi, "Towards Improving Business Processes based on preconfigured KPI target values, Process Mining and Redesign Patterns," *Procedia Computer Science*, vol. 164, pp. 279–284, 2019, doi: 10.1016/j.procs.2019.12.184.
- [29] M. Dumas, M. La Rosa, J. Mendling, and H. A. Reijers, *Fundamentals of Business Process Management*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2018. doi: 10.1007/978-3-662-56509-4.